Physical Chemistry Cume Kinetics Sept 27 2019

1) For the following parallel elementary reactions

$$A \xrightarrow{k_1} B$$
$$A \xrightarrow{k_2} C$$
$$A \xrightarrow{k_3} \mathrm{D}$$

a) Starting with an initial concentration of A = $[A]_0$, obtain an integrated rate equation for [A] as a function of time in terms of the elementary rate constants (30 points)

b) Assuming that the concentrations of B, C and D are initially zero, and can each be separately determined at a given time point, show how one is able to determine the specific values of k_1, k_2 and k_3. (30 points)

2) For the following sequence of reactions

$$\mathrm{A} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{B} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} \mathrm{C} \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} \mathrm{D}$$

Assuming that the concentration of A is maintained at a constant value and that D is removed as soon as it is formed, use the steady state approximation to derive the equation for the rate of formation of D in terms of [A] and the elementary rate constants. (40 points)